

KANTIPUR ENGINEERING COLLEGE

(Affiliated to Tribhuvan University, IOE)

DHAPAKHEL, LALITPUR



[SUBJECT CODE: CT654]

A MINOR PROJECT MID-TERM REPORT ON

MUHAAR : A FACIAL SKIN TYPE DETECTION TOOL

SUBMITTED BY:

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Swati Mahato	KAN079BCT085

SUBMITTED TO:

**DEPARTMENT OF COMPUTER AND ELECTRONICS
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**IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR
THE DEGREE OF BACHELOR IN COMPUTER ENGINEERING**

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LIST OF ABBREVIATIONS

Adam	– Adaptive Moment Estimation (optimizer)
AI	– Artificial Intelligence
BSTI	– Baumann Skin Type Indicator
CNN	– Convolutional Neural Network
DO	– Dry/Oily
FP	– False Positive
FN	–False Negative
GAP	– Global Average Pooling
Google Collab	– Google Collaboratory
GPU	– Graphics Processing Unit
IDE	– Integrated Development Environment
LSTM	– Long Short-Term Memory
NumPy	– Numerical Python
OpenCV	– Open-Source Computer Vision Library
OS	– Operating System
PN	– Pigmented/Non-pigmented
RAM	– Random Access Memory
ReLU	– Rectified Linear Unit (activation function)
RMSProp	– Root Mean Square Propagation (optimizer)
ROI	– Region of Interest
SGD	– Stochastic Gradient Descent (optimizer)
SR	– Sensitive/Resistant
SSD	– Solid State Drive
TP	–True Positive
TN	–True Negative

UV – Ultraviolet

WT – Wrinkle-prone/Tight

CHAPTER 1

INTRODUCTION

1.1. Background

Skin is the biggest organ of the body and one of the most crucial organs to guard us against the threats of the environment, such as pollution, toxic UV light, and bacteria. It also helps in regulating body temperature, preventing dehydration, and serving as a protective cover between our inner system and outer world. Since it's so exposed, our skin is susceptible to several conditions such as acne, eczema, premature aging, or dryness, most of which are dependent on our skin type[1],[3].

Choosing the perfect skincare products depends largely on knowing your skin type. For example, individuals with dry skin would require products containing moisturizing agents such as hyaluronic acid, glycerin, and ceramides to restore lost moisture and strengthen the skin barrier. Individuals with oily skin, on the contrary, require light, oil-free products that can control excess oil and prevent pore blockages such as salicylic acid or niacinamide[3].

This is where skin type detection comes into play. Through exactly identifying whether one has dry, oily, or combination skin, skin care products and routines can then be personalized to meet the skin's individual needs. Computerized detection systems, powered by computer vision and AI, allow this to be processed at a quicker pace, with convenience, and in accessibility, especially for those who may not be able to take advantage of dermatological consultation[4]. Ultimately, skin type recognition allows individuals to make educated decisions about their skincare, leading to better results and improved confidence in their daily practice[5].

1.2. Problem Statement

The most exposed and sensitive part of skin is the face, most people continue to ignore its well-being and health. The daily routine subjects' skin to pollution, harsh climate, stress, and dehydration, but individuals barely pay attention to how these factors affect their facial skin. This absence of information is potentially causing widespread but possibly serious issues such as acne, dryness, premature aging, infection, and irregular skin tone. One of the root causes of this growing phenomenon is the lack of information about one's own skin type and how it reacts to changes in

the environment. Since one does not know what specific needs their skin has, they are unable to take the right steps towards its health, only realizing the damage when problems are apparent or aggravating. Ignorance not only impacts how one looks, but it can also ruin self-confidence and overall well-being.

Therefore, a system that provides knowledge about basic skin type is necessary and crucial for preventing long-term skin issues and encouraging healthy skin.

1.3. Objective

- i. To develop a model that classifies the human skin type in categories like dry, normal and oily.
- ii. To reduce dependency on visual guess work for skin type analysis which is often inaccurate.

1.4. Application and Scope

- i. It can be used various skin care app for skin care recommendations
- ii. It acts as a dermatology virtual assistant to pre assess skin type
- iii. It can be used by professionals before facials, peels, or laser treatments for spa.
- iv. It can be used by dermatology students who are learning in skincare-related fields by providing data.
- v. It allows influencers to use image-based analysis to validate product use cases

1.5. Features

- i. Accurately detect the skin type as dry, normal, oily
- ii. Eliminates the need for physical tests instead users can simply upload a photo for skin type detection
- iii. Designed with modularity to allow future feature upgrades like acne detection.
- iv. It is designed in way that it can easily integrate into web or mobile apps related to skin care

1.6. Feasibility Study

Before implementing the project design, conducting a feasibility analysis was essential. This analysis provided insight on how the project will perform and its potential impact in real-world scenarios, making it a crucial step before proceeding further.

1.6.1. Economic Feasibility

This determines whether the doing of the project is economically beneficial or not. Development will be carried out using existing hardware and standard workstations, avoiding any additional equipment expenses. This will significantly reduce the overall budget required for implementation. The cost-effectiveness of the chosen tools ensures that the system remains accessible and affordable for future development. Moreover, updates and improvements will be integrated over time without imposing extra financial strain, making the solution sustainable in the long run.

1.6.2. Technical Feasibility

The project will be technically feasible because the necessary software and tools such as: Vs Code, JavaScript, TensorFlow, Keras etc. will be readily available and could be downloaded from various online resources at no cost. Also, this system can adopt technological upgrades as it is developed under the consideration of software engineering principles.

1.6.3. Operational Feasibility

The system is user-friendly and compatible with standard phone or laptop cameras, requiring no special equipment. It offers an easy setup where users can simply upload or capture a photo to receive immediate results. Designed for broad accessibility, the project serves general users, skincare enthusiasts, and professionals. The system is scalable, with the potential to expand its capabilities to detect various skin issues. It delivers accurate and reliable analysis through advanced image processing techniques

1.6.4. Schedule Feasibility

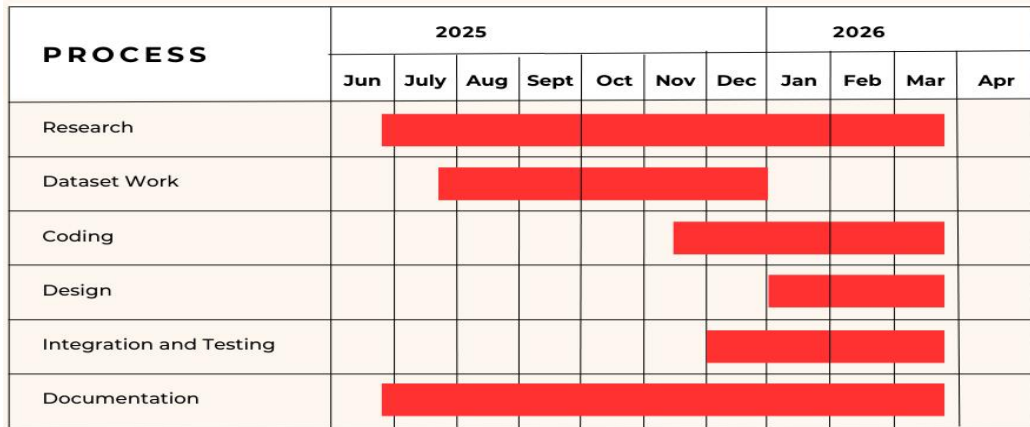


Fig 1.1: Gantt Chart

The project followed a team-based workload distribution, with tasks strategically divided among members. An incremental development approach was adopted, allowing the system to be built into module by module. Each member was responsible for specific components aligned with the project timeline. Regular progress tracking and check-ins ensured the project stayed on schedule, resulting in the timely completion of all major deliverables.

1.7. System Requirements

1.7.1. Development Requirements

Table 1.1: Hardware Requirements

Components	Minimum Requirements
Processor	Intel i5 7 th Generation
RAM	8 GB
Storage	256 GB SSD
GPU	Intel UHD/AMD Vega

Table 1.2: Software Requirements

Components	Minimum Requirements
Operating System	Windows 10, MacOS, Linux
Language	Python, JavaScript
Design Tools	Draw.io, Canva

ML libraries	TensorFlow, Keras, NumPy
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1.7.2. Deployment Requirements

Table 1.3: Hardware Requirements

Components	Minimum Requirements
Processor	Dual Core (2 vCPU)
RAM	2-4 GB
Storage	10-20 GB SSD

Table 1.4: Software Requirements

Components	Minimum Requirements
Operating System	Windows 7, MacOS, Linux
Internet	1-2Mbps

CHAPTER 2

LITERATURE REVIEW

2.1. Related Search

The paper by Jianghong Ran et al. [1] proposes an automatic skin type measurement system based on an improved Inception-v3 deep learning model with transfer learning. The system classifies skin types using the four-dimensional BSTI framework such as dry/oily (DO), sensitive/resistant (SR), pigmented/non pigmented (PN), and wrinkle-prone/tight (WT). Training was performed on a multimodal dataset from 54 healthy Chinese volunteers, including physiological skin indicators, BSTI questionnaire responses, and facial images taken under three lighting conditions. Images were preprocessed using Gaussian filtering and region-of-interest extraction to enhance relevant features. The model achieved high accuracy: 91.11% (DO), 81.13% (SR), 91.72% (PN), and 74.9% (WT), surpassing traditional and previous deep learning methods. However, limitations include the small, demographically narrow dataset and potential variability from lighting conditions. Future improvements should focus on expanding the dataset of diverse populations. This method provides a more objective and efficient alternative for skin type assessment in dermatology and cosmetics.

The paper by Ruchika Chouhan et al. [2] presents a deep learning approach for classifying human skin types to improve cosmetic product recommendations. The study utilizes a substantial dataset of 3,152 frontal facial images of individuals aged 20 to 60, categorized into normal (1,274 images), oily (1,120 images), and dry (758 images) skin types, and further augmented through techniques such as rotation, scaling, flipping, and color adjustment to expand the dataset to 3,408 images. Several well-known transfers learning models, including AlexNet, VGG16, and ResNet50, were evaluated, but the best results were achieved by a custom fine-tuned CNN architecture model features a structured image processing pipeline: images are imported from various devices, preprocessed and enhanced, and then classified using the CNN, which extracts the features for accurate skin type identification. The system not only classifies skin types but is also designed to recommend suitable cosmetic products based on the classification. The fine-tuned CNN achieved a validation accuracy of 99.62% with an average loss of 22.74%, and after further hyperparameter tuning, the accuracy reached 94.57% with a validation loss of 0.0113, demonstrating

the model's high effectiveness and potential for real-world cosmetic and dermatological applications.

The study by Sirawit Saiwaeo et al. [3] proposed a fine-tuned CNN architecture specifically designed for human skin type classification, categorizing skin into normal, oily, and dry types. Initially, well-known pre-trained models such as AlexNet, VGG16, and ResNet50 were evaluated using transfer learning but showed suboptimal performance for this task. The researchers compiled a dataset of 3,152 high-quality frontal facial images from individuals aged 20 to 60, with balanced representation across the three skin types. This dataset was split into training, validation, and testing subsets and augmented with techniques including rotation, scaling, flipping, and color adjustment to improve model robustness. During evaluation, using a new unseen dataset reported an accuracy of 89.70% (21.68% loss). Despite these promising results, the model has limitations such as potential overfitting due to the relatively modest dataset size and limited diversity, as well as challenges in generalizing to skin types or conditions not represented in the training data.

Fatma et al. [4] with other researchers who examine deep learning methods for skin type classification using clinical information. Compare CNN and LSTM models with hyperparameters being optimized as activation functions (Tanh, ReLU, Sigmoid), optimizers (Adam, SGD, RMSProp), and training epochs (50, 100, 500). Data was split between training (7,887), validation (1,126), and testing (2,253). CNN using Adam optimizer, Sigmoid activation, and 500 epochs performed the best with 83.75% accuracy, beating LSTM and other configurations, showing CNN's capability for this nonlinear classification task. Limitations, however, are that there is a single data source, lack of model interpretability analysis, and no comparison with dermatologists.

This research presents an AI-powered system for personalized skincare recommendations by combining ingredient analysis and facial skin assessment. Jinhee Lee et al. [5] uses a Transformer encoder to analyze cosmetic ingredient sequences and predict effectiveness for six skin concerns, while a modified U-Net processes facial images to detect pores, redness, acne, wrinkles, pigmentation, and dark circles. The system was trained on 8,000 curated cosmetic products with ingredient lists and mapped marketing tags, and annotated facial images from the LUMINI Kiosk V2. Results showed strong performance in matching products to skin concerns. However,

limitations include unknown ingredient quantities, a relatively small dataset. Overall, the study shows promise for AI-driven skincare, but emphasizes the need for richer, more diverse data to improve accuracy and reliability.

CHAPTER 3

METHODOLOGY

3.1 Model Training Process

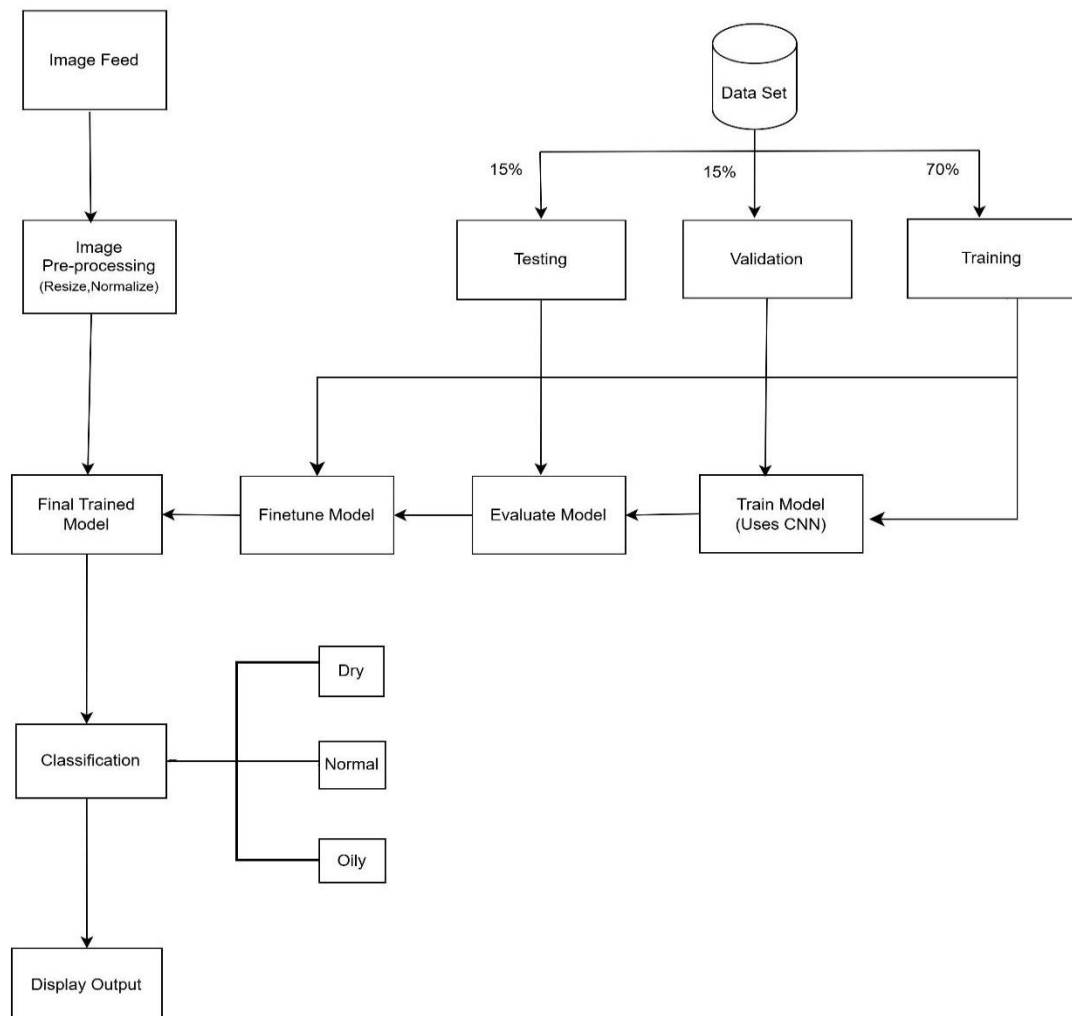


Figure 3.1. Model Training Diagram

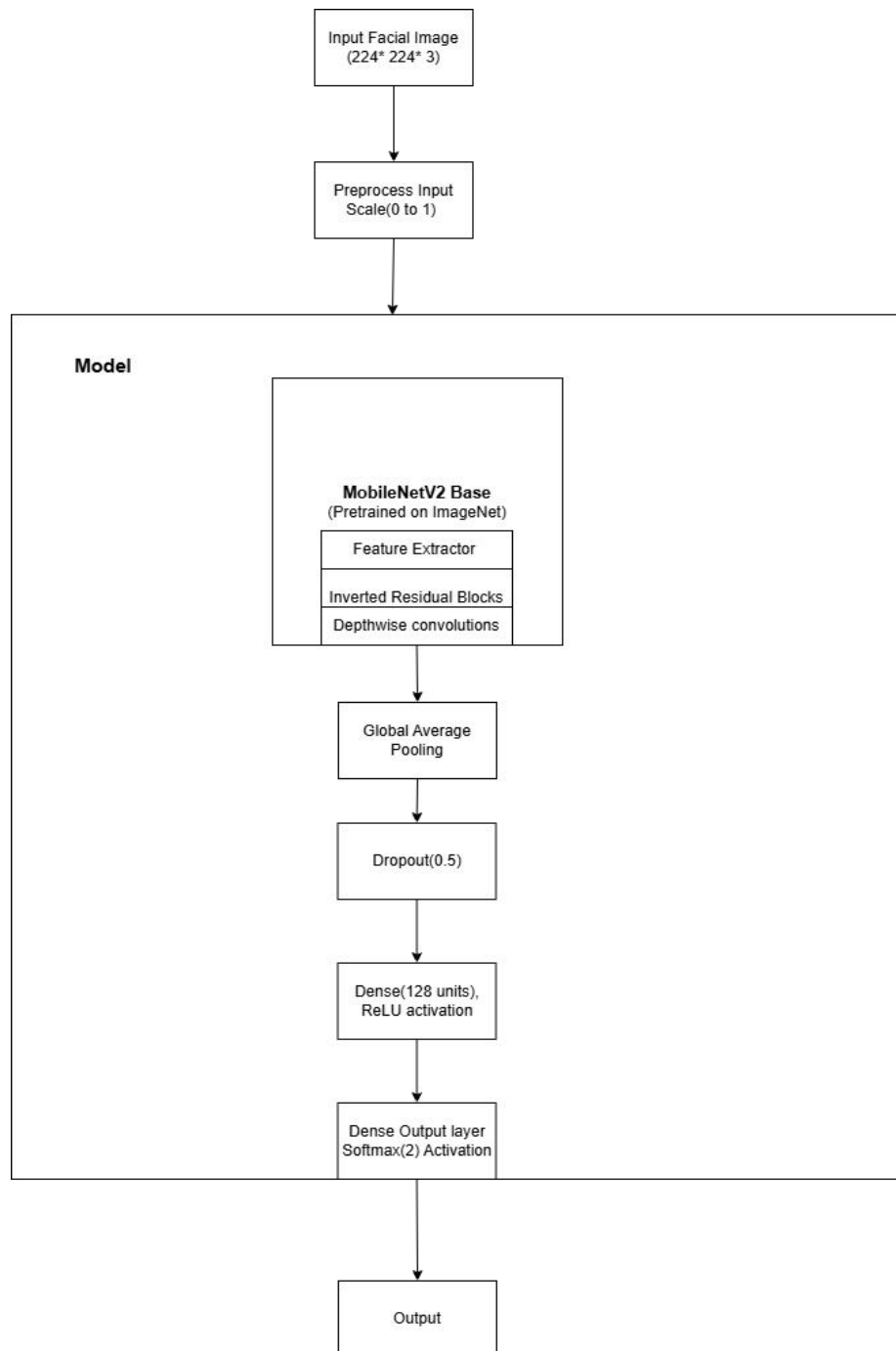


Figure 3.2: MobileNetv2 Training Diagram

3.1.1. Components of Model Training Diagram

1. Data Set Collection

The dataset for skin type detection are gathered from various sources, including publicly available datasets and from Kaggle, which offers datasets specifically labeled for oily, dry, and normal skin types. The dataset will act as the "textbook" for the model, providing examples from which it will learn to recognize skin types. Large, diverse, and well-labeled datasets will enable the model to generalize well beyond the training data, reducing errors and improving performance on real world inputs.

2. Data Pre-processing

i. Data Cleaning

The collected facial skin images were cleaned by removing unclear and irrelevant images. Through this process, better quality and relevant images were retained for training the model, which helped in learning accurate skin texture and features.

ii. Image Resizing:

All the cropped facial images were resized to a fixed dimension of 224×224 pixels to satisfy the input requirements of the MobileNetV2 model. This ensured uniformity across the dataset and enabled efficient processing during model training.

iii. Normalization:

To improve training efficiency, the image pixel values were normalized by scaling them from the range of 0–255 to 0–1. This helped stabilize the learning process and improve the convergence speed of the neural network.

iv. Data Augmentation:

Since the available dataset was limited, data augmentation techniques such as rotation, horizontal flipping, and zooming were applied. These transformations increased data diversity, reduced overfitting, and improved the model's ability to generalize unseen facial images.

v. **Data Splitting:**

Finally, the preprocessed dataset was split into training, validation, and testing sets in an appropriate ratio. This allowed effective model training, performance tuning during validation, and unbiased evaluation of unseen data.

3.2. Overview of the System

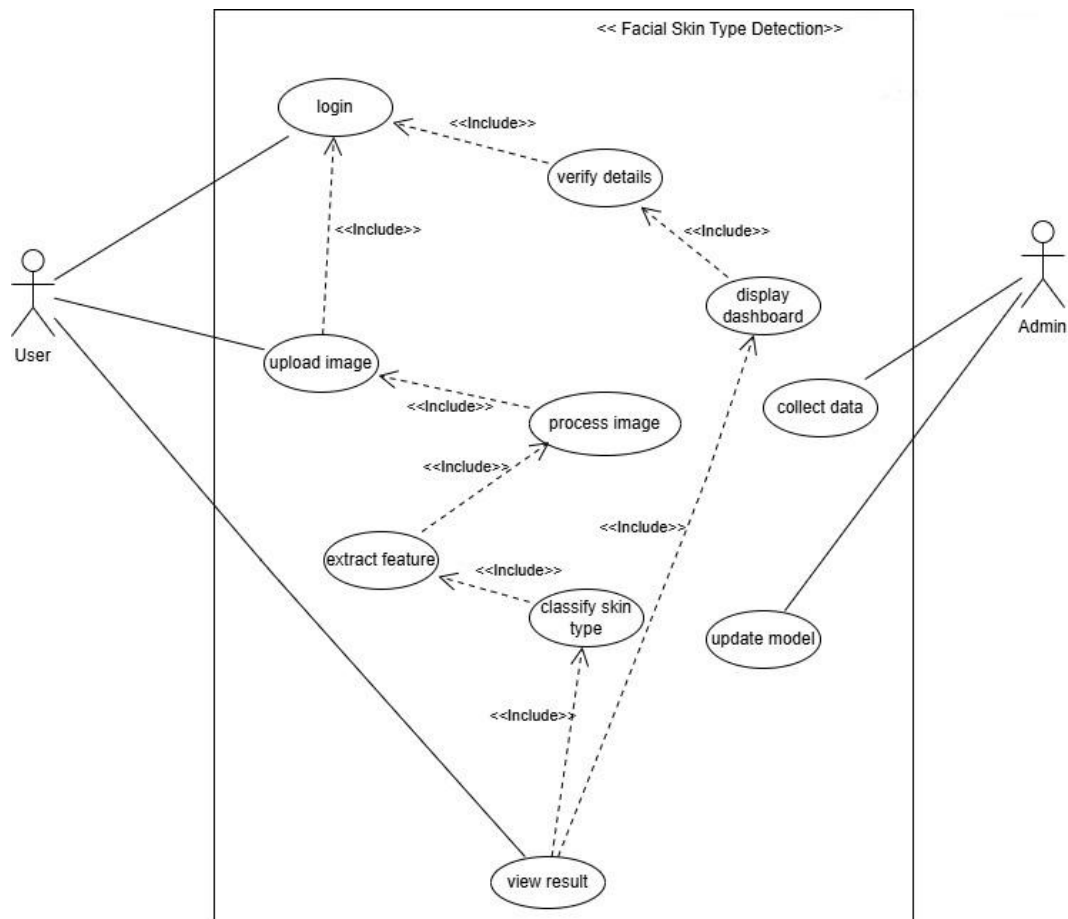


Figure 3.3: Use-Case Diagram

3.3. Software Development Life Cycle

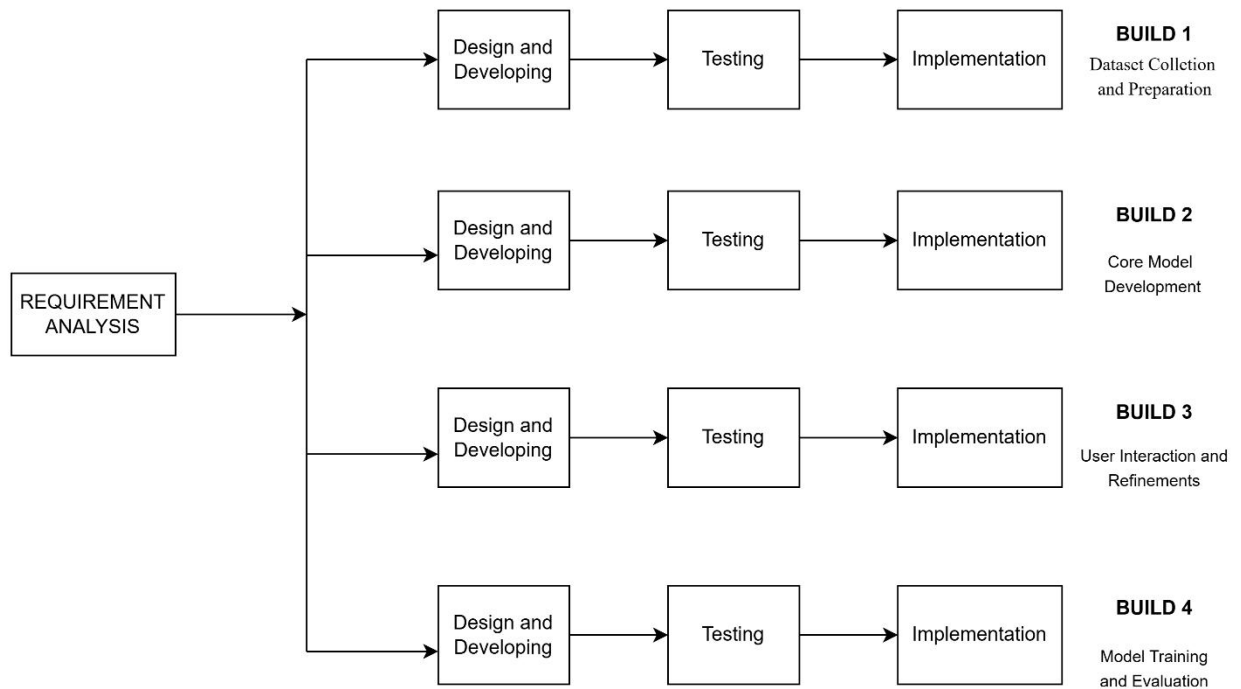


Figure 3.4: Software Development Lifecycle

CHAPTER 4

THEORETICAL FOUNDATION

4.1 Model Architecture

The proposed model architecture is based on a CNN using transfer learning with the MobileNetV2 architecture. The model is designed to automatically extract discriminative facial skin features from images and classify them into three categories: dry, normal, and oily. So basically, the architecture consists of two main components, namely as a feature extractor and a classifier.

- i. **Input Layer:** Each facial image is preprocessed by resizing it to 224×224 pixels with three RGB channels and normalizing the pixel values. This ensures consistent input size and scale, enabling MobileNetV2 to process facial features efficiently and accurately.
- ii. **Feature Extraction Module:** The model uses MobileNetV2 as a feature extractor, a lightweight CNN pre-trained on the ImageNet dataset. The base layers are frozen to retain learned knowledge like edges and lines and reduce overfitting on the small dataset. MobileNetV2's depth-wise separable convolutions ensure low computational cost while maintaining strong feature extraction, making it well suited for facial skin type analysis.
- iii. **MobileNetV2 Architecture:** MobileNetV2 is a lightweight and efficient CNN pre-trained on the ImageNet dataset. It uses depth-wise separable convolutions to reduce computational cost and model size while maintaining strong feature extraction performance.

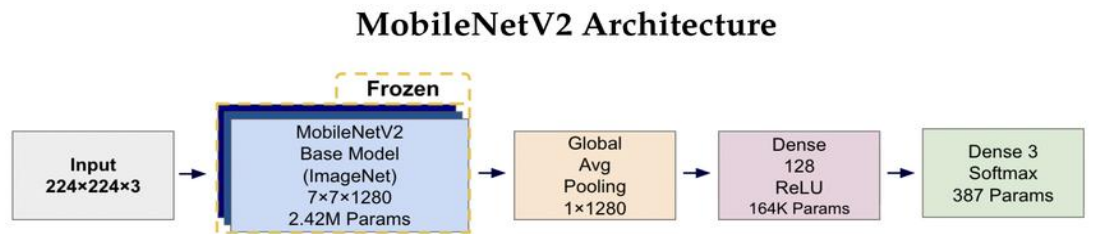


Fig 4.1: MobileNetV2 Architecture

Source: <https://share.google/mvJUGo6WD5tQEDDeI>

- iv. **Depth-wise Separable Convolution:** Depth-wise separable convolution in MobileNetV2 splits standard convolution into two steps: depth-wise convolution

extracts spatial features from each channel, and pointwise (1×1) convolution combines information across channels. This reduces computations and parameters, speeds up the model, and helps prevent overfitting, especially with small datasets.

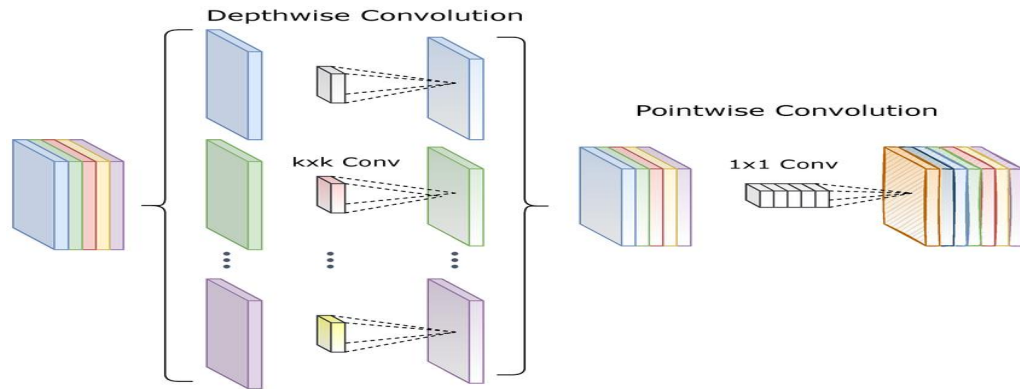


Fig 4.2: Depth wise Convolution and Pointwise Convolution.

Source: <https://share.google/O9tYc4oFK2VYo1IkQ>

- v. **Freezing the Base Model:** Freezing the base model keeps MobileNetV2's pre-trained weights unchanged, preserving learned features like edges and textures. This allows the model to focus on training only the new skin-type classification layers, reducing overfitting, and improving training efficiency on a small dataset.
- vi. **Global Average Pooling layer:** After feature extraction, a Global Average Pooling (GAP) layer is used to reduce high-dimensional feature maps. GAP averages each feature map into a single value, producing a compact 1D feature vector that lowers model complexity, reduces overfitting, and improves classification efficiency.
- vii. **Classification Module:** The classification module maps feature from MobileNetV2 to skin types (dry, normal, oily). The extracted feature maps are flattened into a 1D vector and passed through a Dense layer with ReLU to learn complex patterns. A Dropout layer reduces overfitting, and the final Dense layer with SoftMax outputs class probabilities, with the highest value giving the predicted skin type.

4. Model Evaluating

To evaluate the model, we feed the testing datasets which is 15% of the entire data which has already been split before. The model is then evaluated based on the evaluation of metrics, mainly test accuracy and testing loss. Furthermore, Classification Report, consisting of other evaluation metrics, are also generated.

Evaluation Matrices:

i. Precision

Precision measures how many of the predicted positive samples are correct.

$$\text{Precision} = \frac{TP}{TP+FP} \quad (4.1)$$

ii. Recall (Sensitivity)

Recall measures how many actual positive samples were correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN} \quad (4.2)$$

iii. F1-Score

F1-score is the harmonic mean of Precision and Recall, providing a balanced measure.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.3)$$

iv. Accuracy

Accuracy measures the overall correctness of the mode

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN} \quad (4.4)$$

v. Confusion Matrix

A confusion matrix summarizes prediction results by comparing actual and predicted classes

Actual/ Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

For multi-class classification (dry, normal, oily), a confusion matrix is extended to a 3×3 matrix, where each row represents the actual class, and each column represents the predicted class

CHAPTER 5

RESULT AND DISCUSSION

4.1 Work Accomplished

4.1.1 Data Collection

The data collection process was completed by gathering facial images representing different skin types, including oily, dry, and normal skin. The dataset was primarily collected from Kaggle, which provided labeled and well-structured facial image data suitable for skin type classification. Balanced samples for each skin type were ensured to support effective and unbiased model training.

4.1.2 Data Cleaning

The data cleaning process was carried out to enhance the quality of the collected dataset. This step was performed manually, where unclear and irrelevant images were carefully reviewed and removed.

4.1.3 Data Normalization

The data normalization process was performed to maintain consistency across all facial images. Pixel values were scaled to a standard range. This process helped stabilize the training process and improved the convergence speed of the model.

4.1.4 Data Augmentation

The data augmentation process was applied to increase the effective size of the datasets. Techniques such as image rotation, flipping, zooming, and shifting were used to generate varied versions of the original images. This step helped improve the model's generalization ability and reduced the risk of over fitting.

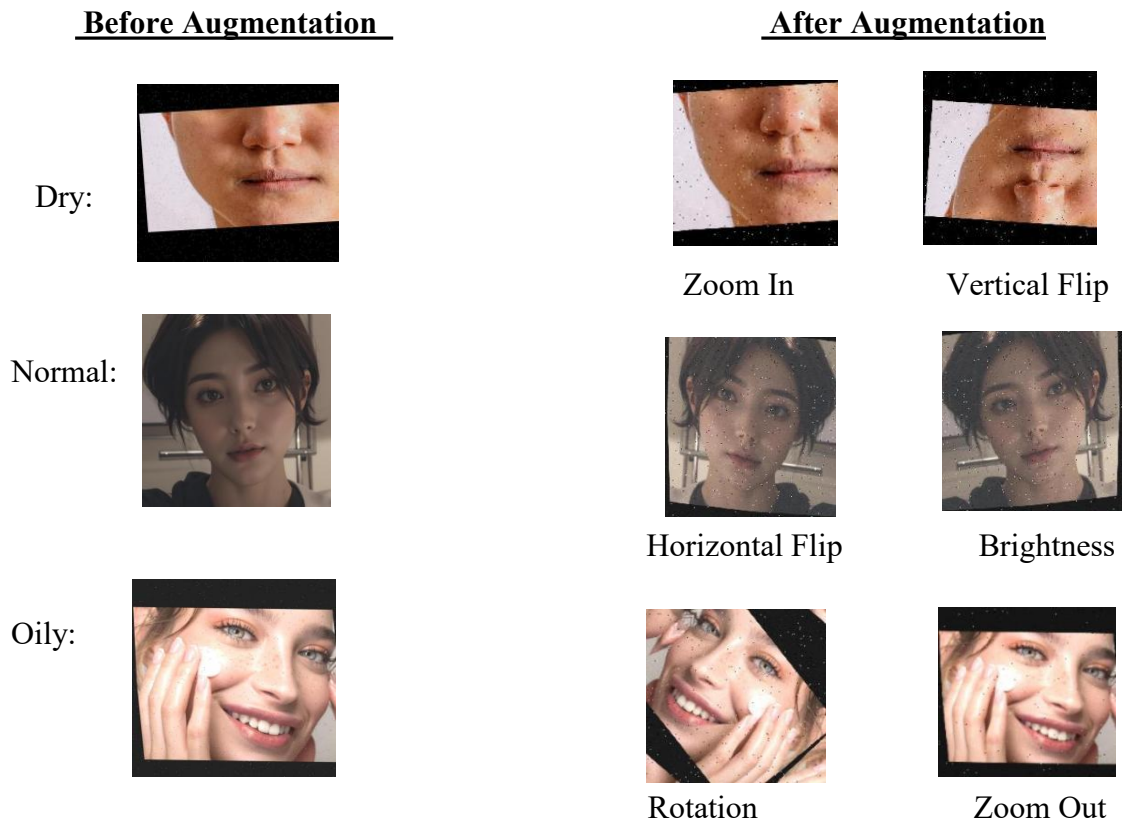


Figure 5.1: Augmented Images

4.1.7 Model Training and Evaluation

The training process for the skin type classification task utilizes a Transfer Learning strategy based on the MobileNetV2 architecture. Initially, the base model layers are frozen to preserve the pre-trained weights from the ImageNet dataset, allowing the newly added classification head to learn task-specific patterns such as skin texture and tone without disturbing general visual features. The final model is evaluated using a hold-out test set to generate a Confusion Matrix and a Classification Report, ensuring the system generalizes well to unseen data.

Validation Loss : 0.4320

Validation Accuracy : 0.8400

Class	Precision	Recall	F1-Score
Dry	0.90	0.91	0.91
Normal	0.93	0.89	0.91
Oily	0.91	0.94	0.93
Accuracy			0.91
Macro Avg	0.91	0.91	0.91

Weighted Avg	0.91	0.91	0.91
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Table 5.1: Classification Report

Confusion Matrix

		<i>Predicted Labels</i>		
		Dry	Normal	Oily
<i>Actual Labels</i>	Dry	373	19	17
	Normal	29	437	26
	Oily	12	15	434

Table 5.2: Confusion Matrix

Observation while training the model

Both the training and validation accuracy increases sharply during the first few epochs i.e from epochs 1 - 7. This indicates that the MobileNetV2 architecture is effectively leveraging its pre-trained ImageNet weights to quickly learn the specific features of "Oily," "Dry," and "Normal" skin types. The curves for training and validation are close to each other which might indicate good generalization. There are some fluctuations in the later training as it is normal during the training process. It suggested that the model is improving and adjusting.

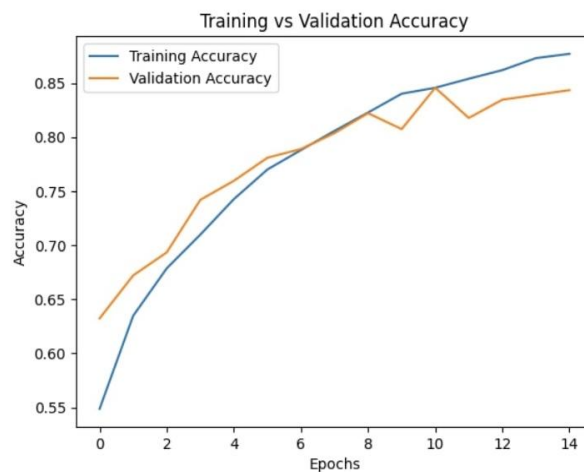


Figure 5.3: Training vs Validation Accuracy

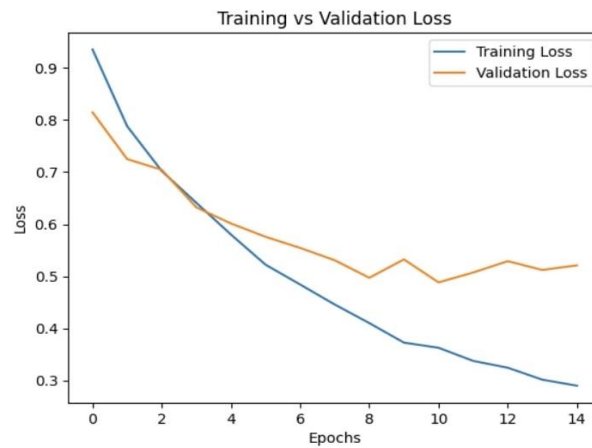


Figure 5.4: Training vs Validation Loss

4.2 Work Remaining

4.2.2 Fine-Tune Model

Fine-tuning is the process of unfreezing the top part of the MobileNetV2 base model and training them so that the pre-trained weights can "fine-tune" themselves to the specific textures of skin types (Oily, Dry, Normal). In our current state, our model has a "frozen" base layers. To complete the fine-tuning, we must unfreeze a portion of the base model, specifically the last few blocks and retrain with a very small learning rate while still benefitting from the pretrained knowledge.

Fine Tuning Steps:

Unfreezing the Top Layers: Instead of unfreezing the entire model (which might destroy the learned ImageNet weights), we will unfreeze only the last 20 to 50 layers. In MobileNetV2, these layers represent the most "complex" features that are most likely to benefit from skin-specific adjustments.

Using a Very Low Learning Rate: This is the most important theoretical step. Since the weights are already very good, a large learning rate would wreck them. It will be the same learning rate used for initial training, but in this case, it will be even lower, so that no drastic weight updates are introduced that will ensure that the training is stable and does not overfit.

Re-compile and Re-train: Once the layers are unfrozen and the learning rate is lowered, we will compile the model again. We will then train for another 10-20 epochs resulting validation accuracy rising higher.

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